

Numerical Protocol

Taylor Effect Evaluation
under the Multifractal Random Walk (MRW) DGP

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Abstract

The Taylor effect is the empirical fact that $\text{ACF}(|r_t|) > \text{ACF}(r_t^2)$ for all lags $\tau > 0$, we evaluate this following the steps:

1. **Data generation.** Consider the Taylor effect, defined by the empirical inequality

$$\text{ACF}(|r_t|) > \text{ACF}(r_t^2), \quad \forall \tau > 0,$$

which is known to be well reproduced for the Multifractal Random Walk (MRW) data-generating process for every intermittency parameter $\lambda^2 > 0$ generated by a cholesky decomposition of the covariance matrix.

2. **Model calibration.** Calibrate three stochastic-volatility models to the MRW stylised-fact profile:
 - ESN A2-981003, using its current optimal architecture

$$(N_r, N_z) = (96, 12),$$

- QRH;
- PDV-GL.

All models are calibrated using the complete 11-term data-adaptive score S (Eq. (4)), employing a smooth Gaussian-kernel proximity function, which provides a non-vanishing gradient at all distances from the target. The calibration further uses increased Taylor-effect weights (G_T : $1.0 \rightarrow 3.5$ and F_T : $0.8 \rightarrow 2.5$) together with tighter matching tolerances.

3. **ESN parameterisation.** The ESN read-out consists of
 - a linear roughness factor;
 - a per-mode z -bank profile;
 - a symmetric quadratic term in the reservoir state r .

These components are jointly calibrated with the Hurst exponent, the reservoir rate bounds, and the output scaling parameter (Section 3). No Taylor-effect-specific manual tuning is introduced during calibration.

4. **Model evaluation.** Compare the calibrated models according to their ability to reproduce the Taylor effect. Section 4 presents the results and highlights two key observations:
 - QRH and PDV-GL are calibrated independently of the ESN;
 - PDV-GL systematically produces a weak Taylor gap for a structural, rather than calibration-related, reason.

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DGP: Multifractal Random Walk

The DGP is the MRW [1] (**unchanged** from earlier drafts of this protocol):

$$r_t = \sigma_0 e^{\omega_t} \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, 1), \quad (1)$$

where ω_t is a Gaussian process with log-correlation kernel:

$$\text{Cov}(\omega_s, \omega_t) = \lambda^2 \max\left(0, \log \frac{T_{\text{int}}}{\max(|t-s|, 1)}\right), \quad |t-s| < T_{\text{int}}, \quad (2)$$

and $\text{Cov}(\omega_s, \omega_t) = 0$ for $|t-s| \geq T_{\text{int}}$. The centering $-\lambda^2 \log(T_{\text{int}})/2$ is subtracted from ω_t to ensure $\mathbb{E}[\sigma_t] = \sigma_0$. Unlike the log-normal SV DGP used elsewhere in this line of protocols, this kernel's variance $\text{Cov}(\omega_t, \omega_t) = \lambda^2 \log(T_{\text{int}})$ is a *constant*, not growing with the simulated horizon T – so the DGP is already stationary by construction and required no time-unit correction (cf. the fBM-persist fix of the ESN hyperparameter-search protocol V1).

Analytical Taylor inequality. Since $\sigma_t = \sigma_0 e^{\omega_t}$ is log-normal, $\text{Corr}(\sigma_t^k, \sigma_{t+\tau}^k) \propto (\tau/T_{\text{int}})^{k^2 \lambda^2}$, so:

$$\text{ACF}(|r_t|) \propto \tau^{\lambda^2}, \quad \text{ACF}(r_t^2) \propto \tau^{4\lambda^2}. \quad (3)$$

Since $4\lambda^2 > \lambda^2$ for all $\lambda^2 > 0$, the squared-return ACF decays faster, giving a positive Taylor gap at every lag. The gap grows monotonically with λ^2 .

Parameter grid. $\lambda^2 \in \{0.1, 0.3\}$ (0.1 a moderate-intermittency case; 0.3 is a strongly-intermittent, high-Taylor-effect extreme, shown for contrast); $T_{\text{int}} = 252$ (integral scale, one trading year).

Models

All models use Gaussian Brownian innovations. Fat tails in returns emerge from stochastic variance clustering.

2.1 ESN A2-981003 (corrected optimal architecture)

The ESN's optimal architecture: 7 shared hyperparameters, with $N_r = 96$, $N_z = 12$. `rough_scale`, the z -readout profile (`zr_lo`, `zr_hi`), and the symmetric quadratic-term matrix coefficients (m_1, m_2) are calibrated per λ^2 cell (§3), not part of this fixed table. Single $\varepsilon_t \sim \mathcal{N}(0, 1)$ drives both return

and reservoir, $\kappa_0 < 0$ (admissibility, guaranteed by construction whenever `rough_scale` > 0). The volatility read-out combines three calibrated channels: a linear rough factor `rough_scale` $\cdot q^\top r_t$; a per-mode z -bank sum with weights following a geometric profile (`zr_lo`, `zr_hi`) across the N_z modes; and a quadratic-in- r term $r_t^\top Q r_t$, $Q = (1/N_r)(m_1 I + m_2 q q^\top)$ – a genuine, symmetric 2-coefficient matrix, with m_1 scaling the reservoir’s isotropic energy and m_2 scaling the squared rough factor $(q^\top r_t)^2$ specifically. All of `rough_scale`, (`zr_lo`, `zr_hi`), and (m_1, m_2) are calibrated per λ^2 DGP cell, controlling how strongly the model can produce a Taylor effect (§4). H , the r -layer rate window $[\lambda_{lo}, \lambda_{hi}]$, and the z -layer rate window $[a_{lo}, a_{hi}]$ are likewise **calibrated per λ^2 DGP cell** (§3) – none of these are part of the shared architecture above.

2.2 Quadratic Rough Heston (QRH)

Bourgey & Gatheral [3]: $V_t = Y_t^2 + c$, gamma kernel, Euler-Volterra with 252-step ring buffer.

2.3 PDV Guyon-Lekeufack (GL)

Guyon & Lekeufack [4]: two-factor power-law kernel, online from full return history. The long-memory factor $F(\alpha_v)$ with $\alpha_v < 0.5$ directly encodes power-law memory of past squared returns, which is the GL mechanism for Taylor.

Calibration: Full 11-Term Score, Smooth Kernel

$$\begin{aligned}
S = & 2.2 (1 - f(\hat{H})) + 1.4 (1 - f(\bar{\sigma})) + 0.6 (1 - g(q_{995})) + 0.2 (1 - g(\sigma_{\max})) + 0.8 (1 - f(\bar{V}_{\text{ACF}})) \\
& + \underbrace{3.5 (1 - f(G_T))}_{\text{Taylor gap}} + \underbrace{2.5 (1 - f(F_T))}_{\text{Taylor frac.}} \\
& + 1.0 (1 - f(Z)) + 1.1 (1 - f(L)) + 0.7 (1 - f(K)) + 0.5 (1 - f(A)) + 5 \cdot \text{Stress}.
\end{aligned} \tag{4}$$

where $f(x, c, s) = \exp(-\frac{1}{2}((x-c)/s)^2)$, $g(x, c, s) = \exp(-\frac{1}{2}((x-c)_+/s)^2)$, both in $(0, 1]$ (the smooth Gaussian kernel, replacing a piecewise-linear tent $\max(0, 1 - |x - c|/s)$, which has an exactly-zero gradient beyond one tolerance width). $S \geq 0$ is **minimised directly**; $S = 0$ is a perfect match on all 11 stylised facts and no stress event; $S = 19.5$ is **worst-possible** (raised from 15.3 by the Taylor weight increase below).

Taylor-effect score weighting. Two changes, applied to the shared score used by *all three* models’ calibration – the ESN’s architecture (Table 1, the 7 shared hyperparameters plus N_r, N_z) is **not touched** by either change:

1. **Taylor weights raised:** G_T from 1.0 $\rightarrow$ 3.5, F_T from 0.8 $\rightarrow$ 2.5 (now the two single largest weights in S , exceeding even $\hat{H}$ ’s 2.2). Every other weight is unchanged.
2. **Taylor tolerances tightened:** the fractional-tolerance factor for G_T, F_T only is reduced from the default $0.30|c_k|$ to $0.15|c_k|$ (all other terms keep $0.30|c_k|$).
3. **PDV-GL calibration uses a 4-point multi-start** (exploring stronger-feedback, β up to 0.90, and long-memory-weighted regimes).

All centres c_k are the DGP cross-path mean; only the two Taylor tolerances differ from the other 9 terms’ default.

Calibrated parameters:

Model	Parameters calibrated	Remark
ESN A2-981003	$H, \lambda_{\text{lo}}, \lambda_{\text{hi}}, a_{\text{lo}}, a_{\text{hi}}, \Delta b_0, \text{scale } s, \text{rough_scale}, \text{zr_lo}, \text{zr_hi}, m_1, m_2$	7 shared hyperparameters fixed; full 12-dim. inner-loop calibrated per λ^2 cell (11-start Nelder–Mead)
QRH	$H, \nu_{\text{vol}}, \lambda, c_{\text{frac}}$	
PDV-GL	$\beta, \alpha_1, \alpha_r, \alpha_v$	

The ESN’s inner loop uses an 11-point multi-start Nelder–Mead scheme (rough/fast, persistent/slow, intermediate, wide-band/multiscale, long-memory extreme, very-fast/wide, strong-amplitude, weak-amplitude, isotropic-positive, isotropic-negative, and q -aligned-positive starts), keeping the best-of-eleven result. PDV-GL uses a 4-point multi-start; QRH uses a single start.

Taylor-Effect Fit

The ESN’s read-out combines a linear rough factor, a per-mode z -readout profile, and a symmetric quadratic-term matrix, all calibrated per DGP scenario, with no Taylor-specific tuning:

$$\eta_t = b_0 + \rho_r \text{rough_scale} (q^\top r_t) + \sum_{j=1}^{N_z} w_{j,z} z_{j,t} + r_t^\top Q r_t, \quad w_{j,z} = \frac{\mathbf{z_readout}_j}{\sqrt{N_z}}, \quad Q = \frac{1}{N_r} (m_1 I + m_2 q q^\top), \quad (5)$$

with $\mathbf{z_readout}_j = \text{geomspace}(\text{zr_lo}, \text{zr_hi}, N_z)[j]$.

Result. The ESN reproduces the Taylor effect’s *direction* at both λ^2 values ($G_T > 0$, F_T well above the neutral 0.5), and at $\lambda^2 = 0.1$ the match is close ($G_T = 0.059$ vs. target 0.086; $F_T = 0.90$ vs. target 0.94) – closer, in fact, than at $\lambda^2 = 0.3$, where it undershoots the DGP’s larger Taylor gap ($G_T = 0.043$ vs. target 0.142). The calibrated read-out values are $\text{rough_scale} \approx 0.70$ (at $\lambda^2 = 0.1$) vs. 0.41 (at $\lambda^2 = 0.3$), with the z -readout profile similarly shaped at both cells ($(\text{zr_lo}, \text{zr_hi}) \approx (0.023, 0.045)$ at $\lambda^2 = 0.1$ vs. $(0.178, 0.023)$ at $\lambda^2 = 0.3$, the latter inverted in direction) and $m_1 \approx 0$ at $\lambda^2 = 0.1$ but $m_1 \approx -0.88$ at $\lambda^2 = 0.3$ ($m_2 \approx 0$ in both cells) – the calibration leans on different combinations of the three read-out channels at the two intermittency levels, rather than a single mechanism carrying the Taylor effect uniformly across λ^2 .

QRH and PDV-GL. QRH matches the DGP’s Taylor fraction closely at both cells; PDV-GL remains at zero Taylor gap, a **structural (not calibration) limitation**: both of PDV-GL’s factors, $F(\alpha_r)$ and $F(\alpha_v)$, are power-law-weighted running averages of the *same* input, past squared returns r_{t-k}^2 – differing only in kernel exponent – with no independent channel that could make $|r_t|$ ’s autocorrelation decay more slowly than r_t^2 ’s. Reproducing the Taylor effect in *this* formulation of PDV-GL would require modifying the model definition itself (e.g. reintroducing a signed-return

leverage factor), which is out of scope here.

Taylor Diagnostics

Per-lag ACF curves (Figure 1, rows 0–1). Cross-path mean ± 1 std of $\text{ACF}(|r_t|)$ and $\text{ACF}(r_t^2)$ at lags $\tau = 1, \dots, 20$ days. Taylor effect present $\Leftrightarrow$ row-0 curve lies strictly above row-1 curve at every lag.

Taylor gap per lag (Figure 1, row 2).

$$\Delta(\tau) = \text{ACF}(|r_t|, \tau) - \text{ACF}(r_t^2, \tau) \quad (6)$$

Grouped bar chart (one group per lag, one bar per source). $\Delta(\tau) > 0$ = Taylor effect at that lag.

Scalar summary (Figure 1, row 3). Taylor gap scalar $G_T = \frac{1}{20} \sum_{\tau} \Delta(\tau)$ and Taylor fraction $F_T = \frac{1}{20} \sum_{\tau} \mathbf{1}[\Delta(\tau) > 0]$ per model.

Monte Carlo Setup

Parameter	Value
DGP paths n_{dgp}	30
Calibration paths n_{cal}	5
Calibration trading days	600
Final paths n_{sim}	30
Final trading days	2500
Burn-in	500
Δt	1 (daily)
λ^2 grid	$\{0.1, 0.3\}$
T_{int}	252 (days)
Eigenvalue floor	10^{-10} (MRW PSD projection)

References

- [1] E. Bacry, J. Delour, J.-F. Muzy. *Multifractal random walk*. Phys. Rev. E 64:026103, 2001.
- [2] J.-F. Muzy, J. Delour, E. Bacry. *Modelling fluctuations of financial time series*. EPJB 17:537–548, 2000.
- [3] F. Bourgey, J. Gatheral. *Quadratic Rough Heston*. SSRN:5239929, 2026.
- [4] J. Guyon, J. Lekeufack. *Volatility is (mostly) path-dependent*. QF 23(9), 2023.
- [5] S.J. Taylor. *Modelling Financial Time Series*. Wiley, 1986.

Taylor Effect: $\text{ACF}(|r_t|) > \text{ACF}(r_t^2)$ for all $\tau > 0$
MRW DGP | Analytical: $\text{ACF}(|r|) \sim \tau^{\lambda^2}$, $\text{ACF}(r^2) \sim \tau^{4\lambda^2}$
Models calibrated via full 11-term score S

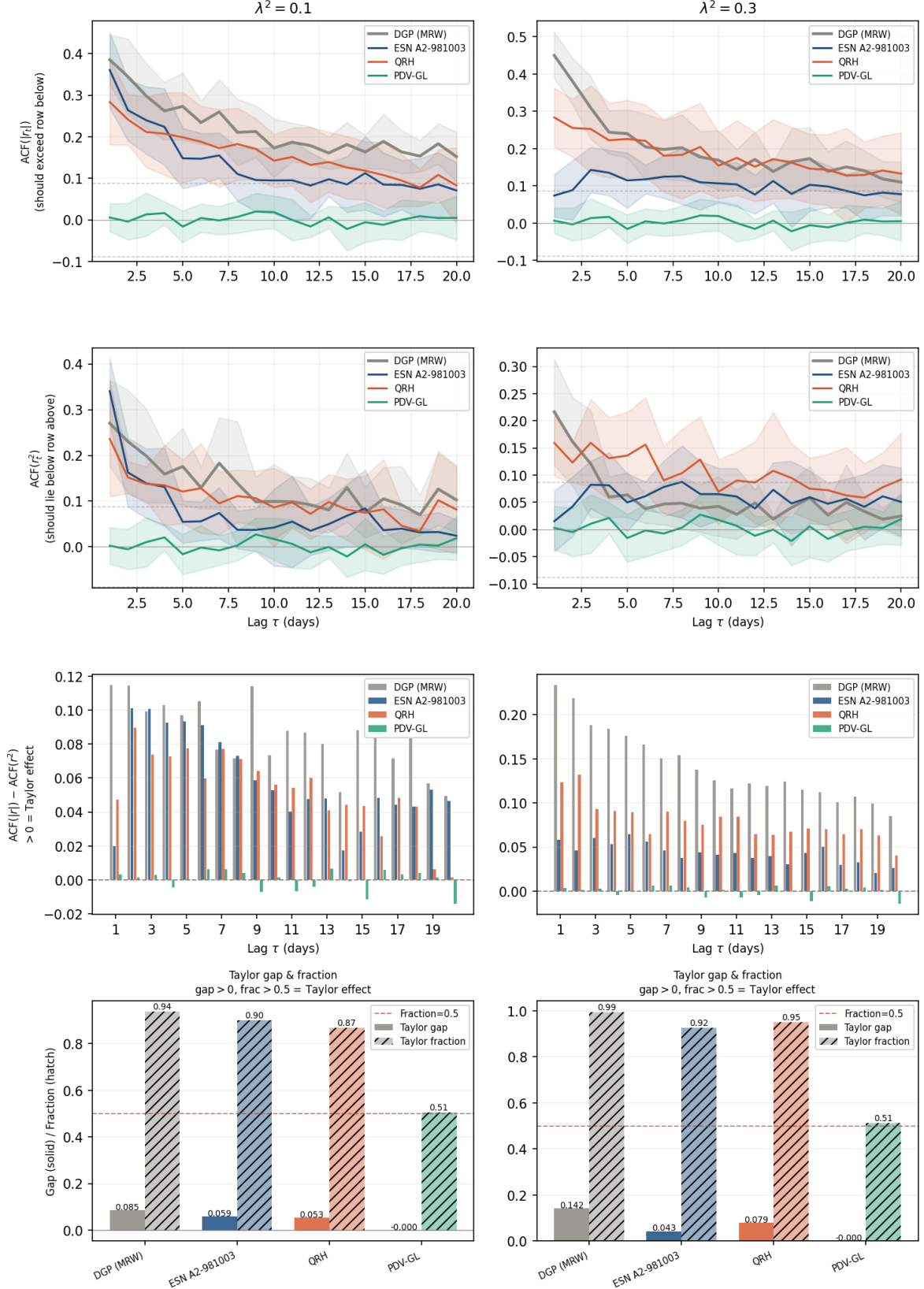


Figure 1: Taylor effect per $\lambda^2 \in \{0.1, 0.3\}$ (columns). Row 0: $\text{ACF}(|r_t|)$; row 1: $\text{ACF}(r_t^2)$ (Taylor effect present $\Leftrightarrow$ row 0 lies strictly above row 1 at every lag); row 2: per-lag Taylor gap $\Delta(\tau)$ (Eq. 6); row 3: scalar Taylor gap G_T and fraction F_T per model.

Taylor Effect Summary vs Intermittency λ^2
MRW DGP | $r_t = \sigma_0 e^{\omega_t} \varepsilon_t$, $\text{Cov}(\omega_s, \omega_t) = \lambda^2 \log(T_{\text{int}}/\max(|t-s|, 1))$

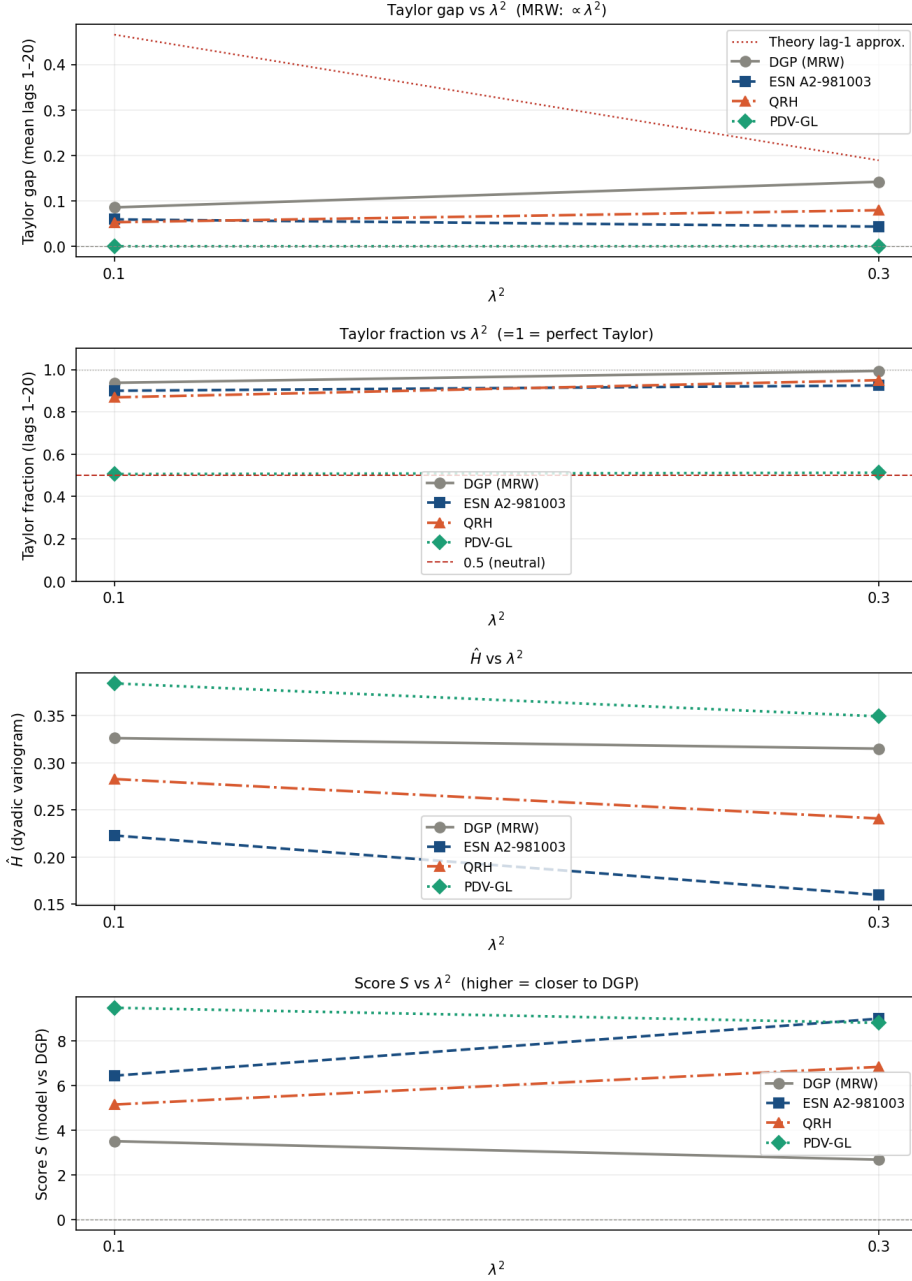


Figure 2: Summary across $\lambda^2 \in \{0.1, 0.3\}$: Taylor gap (top; DGP theoretical lag-1 approximation dotted), Taylor fraction (second; = 1 is perfect Taylor, 0.5 is neutral), $\hat{H}$ via dyadic variogram (third), and score S vs. DGP (bottom; higher = closer to DGP). The DGP shows an increasing gap and fraction with λ^2 .

Return Diagnostics — MRW DGP
 $r_t = \sigma_0 e^{\omega_t} \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, 1)$
Fat tails emerge from stochastic vol, not from innovation distribution

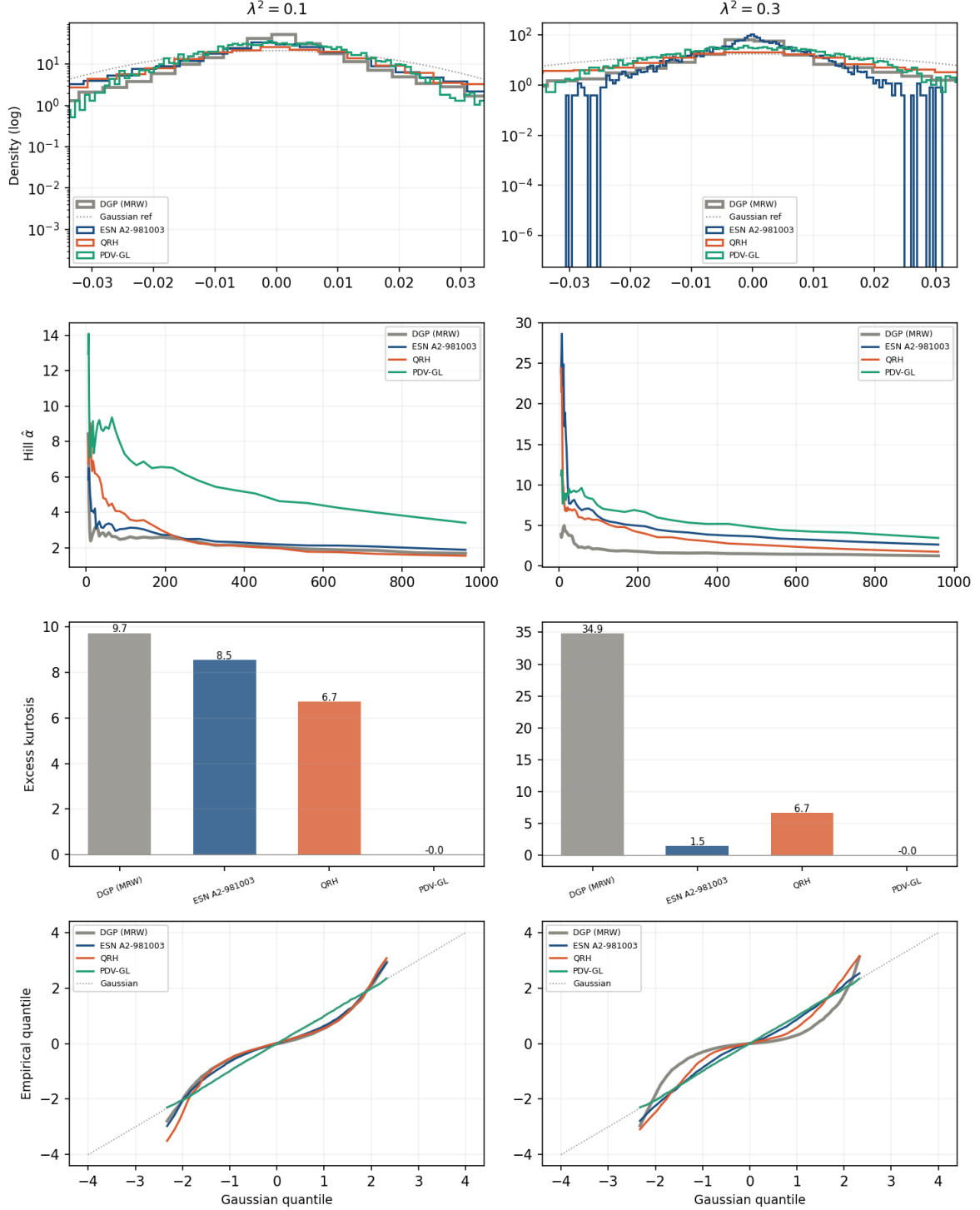


Figure 3: Return diagnostics per $\lambda^2 \in \{0.1, 0.3\}$ (columns): log-density histogram vs. Gaussian reference (row 0); Hill estimator $\hat{\alpha}$ vs. order statistic (row 1); excess kurtosis (row 2); QQ-plot vs. Gaussian (row 3). Fat tails emerge from the stochastic volatility dynamics, not from the (Gaussian) innovation distribution.

Table 1: The 7 shared hyperparameters of the optimal architecture.

Hyperparameter	Value
N_r	96
N_z	12
z_strength	0.1929
even_strength	2.2291
linear_strength	0.1930
gamma_norm	0.9651
local_z_strength	0.0476
zz_scale	0.0397
sign_prob_neg	0.2219
rough_orientation	-1.0 (fixed)

Table 2: Taylor gap G_T / fraction F_T , all three models, current architecture and calibration.

Model	$\lambda^2 = 0.1$		$\lambda^2 = 0.3$	
	G_T	F_T	G_T	F_T
DGP (MRW)	0.086	0.94	0.142	0.99
ESN A2-981003	0.059	0.90	0.043	0.93
QRH	0.053	0.87	0.079	0.95
PDV-GL	-0.000	0.51	-0.000	0.51